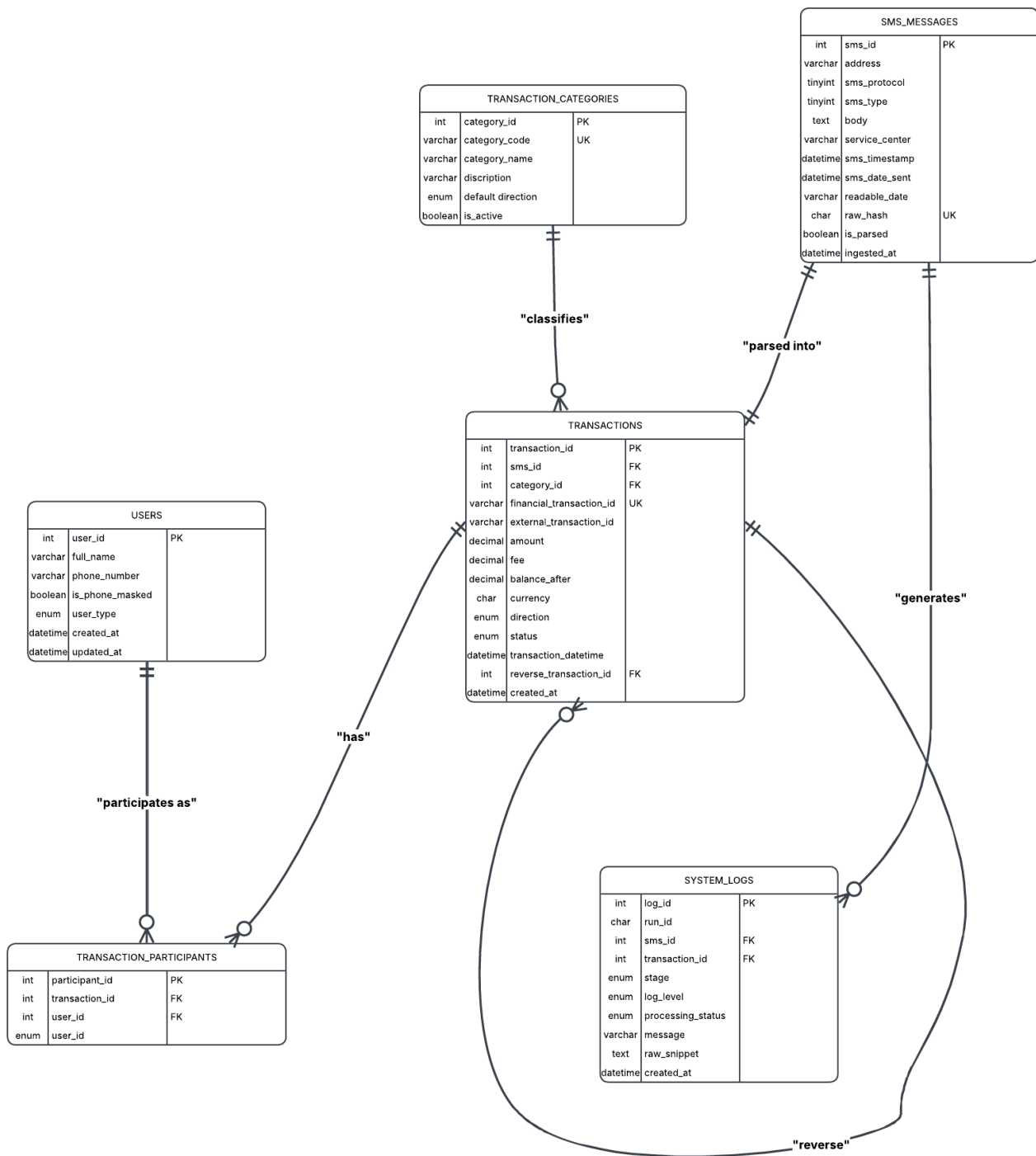


Database Design Document — MoMo SMS Data Processing System

Week 2 deliverable · Urban Mobility Data Explorer (MoMo ETL) team

- ERD source (editable): [erd_diagram.drawio](#) — open at [app.diagrams.net](#)
- ERD image: [erd_diagram.png](#)
- SQL implementation: [../database/database_setup.sql](#)
- JSON serialization examples: [../examples/json_schemas.json](#)

1. Entity Relationship Diagram



Six entities: users, transaction_categories, sms_messages, transactions, transaction_participants, system_logs.

Relationship	Cardinality	Notes
transaction_categories → transactions	1:M	Every transaction has exactly one category
sms_messages → transactions	1:0..1	A message parses into at most one transaction (sms_id is UNIQUE on transactions)

Relationship	Cardinality	Notes
transactions → transaction_participants	1:M	A transaction has 1-3 participant rows
users → transaction_participants	1:M	A user participates in many transactions over time
users ↔ transactions	M:N , resolved by transaction_participants	See §2
transactions → transactions (self)	1:0..1	reverses_transaction_id, set only when status = REVERSED
sms_messages / transactions → system_logs	1:M	ETL audit trail, both FKs nullable

2. Design Rationale (why the schema looks like this)

The schema was derived directly from data/raw/momo.xml, not from a generic template — every design decision below traces back to a specific message pattern found in that file.

Why a junction table instead of sender_id/receiver_id columns. The eleven distinct SMS shapes in the data don't all describe a simple two-party transfer. A bank deposit names no counterparty at all; a merchant payment names one; a peer transfer names one; and an **agent withdrawal** ("You Account Holder... have via agent: Agent John (...), withdrawn...") names two — the account holder and the agent — for a single transaction. Two fixed FK columns can't represent that variability cleanly, so transaction_participants resolves users ↔ transactions as a real many-to-many relationship, with a role (SENDER/RECEIVER/AGENT/ACCOUNT_HOLDER) on each edge. This is the required M:N relationship, and it is evidenced by the source data rather than added just to satisfy the rubric.

Why phone_number is nullable and not unique. The same real-world contact appears both masked (*****013) and in full (250791666666) across different messages, and bank deposits carry only a settlement account number, not a phone number. A UNIQUE constraint on phone would break the first time a masked and unmasked number collided or a deposit arrived with no number at all, so users uses a surrogate user_id and treats phone as an informational, best-effort field.

Why status and the self-referencing reverses_transaction_id. The corpus contains not just completed transactions but explicit failures ("...has failed at...") and reversals ("A reversal has been initiated..."). Modeling only "happy path" transactions would silently drop real data the ETL pipeline has to handle, so status is a first-class ENUM and reversals link back to the original transaction via a nullable self-FK rather than a separate table.

Why sms_messages is a separate table from transactions. Keeping the raw SMS verbatim (1:1 with its parsed transaction via a UNIQUE FK) preserves an audit trail back to the original text and lets system_logs record processing outcomes — including dead-letter entries — for messages that never produce a transaction at all (transaction_id is nullable there for exactly that reason).

Why financial_transaction_id is UNIQUE but nullable. It is MoMo's own transaction identifier (TxId / "Financial Transaction Id" in the SMS body) and doubles as the ETL's idempotency key — re-ingesting the same backup file must not create duplicate rows. It's nullable because a handful of message types (bank deposits, mobile transfers) don't surface one.

3. Data Dictionary

users

Column	Type	Constraints	Description
user_id	INT	PK, AUTO_INCREMENT	Surrogate key
full_name	VARCHAR(100)	NOT NULL	Name as it appears in the SMS body
phone_number	VARCHAR(15)	NULL	May be masked (*****013), full, or absent
is_phone_masked	BOOLEAN	NOT NULL, DEFAULT FALSE	True when phone_number contains * characters
user_type	ENUM	NOT NULL, DEFAULT 'CUSTOMER'	CUSTOMER / AGENT / MERCHANT / SYSTEM
created_at, updated_at	DATETIME	NOT NULL	Row bookkeeping

transaction_categories

Column	Type	Constraints	Description
category_id	INT	PK, AUTO_INCREMENT	Surrogate key
category_code	VARCHAR(30)	NOT NULL, UNIQUE	Stable key used by etl/categorize.py
category_name	VARCHAR(100)	NOT NULL	Dashboard label
description	VARCHAR(255)	NULL	
default_direction	ENUM('CREDIT','DEBIT')	NOT NULL	Typical balance effect

sms_messages

Column	Type	Constraints	Description
sms_id	INT	PK, AUTO_INCREMENT	Surrogate key
address	VARCHAR(30)	NOT NULL	e.g. "M-Money"
body	TEXT	NOT NULL	Raw, unparsed SMS text
sms_timestamp	DATETIME	NOT NULL	Parsed from the date epoch-ms attribute
raw_hash	CHAR(64)	NOT NULL, UNIQUE	SHA-256 of the raw element; de-dupes re-ingested backups
is_parsed	BOOLEAN	NOT NULL, DEFAULT FALSE	Set once a transactions row exists

Column	Type	Constraints	Description
service_center, readable_date, sms_protocol, sms_type, sms_date_sent, ingested_at	—	NULL	Secondary/debug fields carried from the XML

transactions

Column	Type	Constraints	Description
transaction_id	INT	PK, AUTO_INCREMENT	Surrogate key
sms_id	INT	FK → sms_messages, UNIQUE, NOT NULL	1:1 provenance link
category_id	INT	FK → transaction_categories, NOT NULL	
financial_transaction_id	VARCHAR(30)	UNIQUE, NULL	MoMo's own TxId; ETL idempotency key
external_transaction_id	VARCHAR(30)	NULL	Present on some third-party debits
amount	DECIMAL(12,2)	NOT NULL, CHECK (amount > 0)	
fee	DECIMAL(12,2)	NOT NULL, DEFAULT 0, CHECK (fee >= 0)	
balance_after	DECIMAL(12,2)	NULL, CHECK (balance_after >= 0)	Not every message reports it
currency	CHAR(3)	NOT NULL, DEFAULT 'RWF'	
direction	ENUM('CREDIT','DEBIT')	NOT NULL	
status	ENUM('COMPLETED','FAILED','PENDING','REVERSED')	NOT NULL, DEFAULT 'COMPLETED'	
transaction_datetime	DATETIME	NOT NULL	Timestamp embedded in the SMS text
reverses_transaction_id	INT	FK → transactions (self), NULL	Set only when status = REVERSED

transaction_participants (junction table)

Column	Type	Constraints	Description
participant_id	INT	PK, AUTO_INCREMENT	Surrogate key

Column	Type	Constraints	Description
transaction_id	INT	FK → transactions, NOT NULL	
user_id	INT	FK → users, NOT NULL	
role	ENUM('SENDER','RECEIVER','') NOT NULL		
—	—	UNIQUE (transaction_id, role)	A transaction has at most one of each role

system_logs

Column	Type	Constraints	Description
log_id	INT	PK, AUTO_INCREMENT	Surrogate key
sms_id	INT	FK → sms_messages, NULL, ON DELETE SET NULL	
transaction_id	INT	FK → transactions, NULL, ON DELETE SET NULL	
stage	ENUM('PARSE','CLEAN','CATEGORIZE') NOT NULL		
log_level	ENUM('INFO','WARNING','ERROR') NOT NULL, DEFAULT 'INFO'		
processing_status	ENUM('SUCCESS','FAILED','SKIPPED') NOT NULL		
message	VARCHAR(500)	NOT NULL	
raw_snippet	TEXT	NULL	Populated for dead-letter entries

4. JSON Serialization Mapping

examples/json_schemas.json gives one example object per table plus two composite shapes:

- `transaction_flat` — a bare transactions row, FK columns only (what `SELECT *` looks like serialized).
- `transaction_detailed` — the **complex nested object** the assignment asks for: a full transaction with its category resolved and every `transaction_participants` row expanded into a nested user object. This is the shape a `GET /transactions/{id}` endpoint (see `api/schemas.py`) or `web/chart_handler.js` would actually consume.
- `agent_withdrawal_example / reversal_example` — show the 3-party M:N case and the self-referencing reversal in JSON form.
- `dashboard_analytics_example` — an aggregate shape (`GROUP BY category`) for chart data.

5. Verified on MySQL 8.0 (db-fiddle.com, MySQL v8)

The full database/database_setup.sql (all 6 CREATE TABLE statements, all sample INSERTS) was run unmodified against a real MySQL 8.0 instance, not just validated locally, because CHECK constraints are silently ignored before MySQL 8.0.16 — a schema that *parses* is not proof it *enforces* anything. Screenshots below are from that run.

5.1 Sample query — joins + the M:N junction resolved correctly

```
SELECT t.transaction_id, t.financial_transaction_id, t.amount, t.direction, t.status,
       tc.category_name,
       GROUP_CONCAT(CONCAT(tp.role, ': ', u.full_name) ORDER BY tp.role SEPARATOR ' | ') AS participants
FROM transactions t
JOIN transaction_categories tc ON tc.category_id = t.category_id
JOIN transaction_participants tp ON tp.transaction_id = t.transaction_id
JOIN users u ON u.user_id = tp.user_id
GROUP BY t.transaction_id
ORDER BY t.transaction_datetime;
```

Result includes, e.g., transaction 9: AGENT:Agent John | ACCOUNT_HOLDER:Account Holder — a single transaction correctly resolved to two distinct users in two distinct roles, proving the junction table works.

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Fiddle Description

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Schema SQL

```
1 -- =====
2 -- MoMo SMS Data Processing System - Database Setup
3 -- Week 2 deliverable: Urban Mobility Data Explorer (MoMo ETL) team
4 --
5 -- Target engine : MySQL 8.0.16+ (required for CHECK constraint
6 -- enforcement;
7 -- on 5.7/8.0.0-8.0.15 CHECK clauses parse but are
8 -- ignored)
9 -- How to run : mysql -u <user> -p < database_setup.sql
10 -- =====
11
12 -- 1. users
13 -- People or entities that appear as a party (sender/receiver/agent)
14 -- in a transaction. Populated from names + phone numbers found in
15 -- the SMS body text.
16 --
17 -- NOTE: momo.xml masks some phone numbers ("*****013") and shows
```

Text to DDL

Query SQL

```
1 -- Full transaction detail joined with category and every participant
2 -- (proves the M:N junction resolves correctly)
3 SELECT t.transaction_id, t.financial_transaction_id, t.amount, t.direction,
4        t.status,
5        tc.category_name,
6        GROUP_CONCAT(CONCAT(tp.role, ': ', u.full_name) ORDER BY tp.role
7        SEPARATOR ' | ') AS participants
8 FROM transactions t
9 JOIN transaction_categories tc ON tc.category_id = t.category_id
10 JOIN transaction_participants tp ON tp.transaction_id = t.transaction_id
11 JOIN users u ON u.user_id = tp.user_id
12 GROUP BY t.transaction_id
13 ORDER BY t.transaction_datetime;
```

Results

Copy as Markdown

2	73214484437	1000.0	DEBIT	COMPLETED	Payment to Code Holder	SENDER:Account Holder RECEIVER:Jane Smith
3	null	40000.0	CREDIT	COMPLETED	Bank Deposit	RECEIVER:Account Holder
4	null	10000.0	DEBIT	COMPLETED	Transfer to Mobile Number	SENDER:Account Holder RECEIVER:Samuel Carter
5	13913173274	2000.0	DEBIT	COMPLETED	Airtime Bill Payment	SENDER:Account Holder
6	14264876273	2000.0	DEBIT	COMPLETED	Cash Power Bill Payment	SENDER:Account Holder
7	14324965479	2000.0	DEBIT	COMPLETED	Bundles and Packs Purchase	SENDER:Account Holder
8	13947831685	25000.0	DEBIT	COMPLETED	Third Party Transaction	SENDER:Account Holder RECEIVER:DIRECT PAYMENT LTD
9	98765432101	15000.0	DEBIT	COMPLETED	Withdrawal from Agent	AGENT:Agent John ACCOUNT_HOLDER:Account Holder

5.2 Basic CRUD operations

Operation	Statement	Verified result
CREATE	INSERT INTO users (...) VALUES ('Grace Uwimana', ...)	grace_uwimana_created = 1
UPDATE	UPDATE transactions SET amount = 1500.00 WHERE transaction_id = 2	txn2_amount_after_update = 1500.0
DELETE	DELETE FROM system_logs WHERE stage='PARSE' AND processing_status='FAILED'	dead_letter_logs_remaining = 0

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14 -- in a transaction. Populated from names + phone numbers found in
15 -- the SMS body text.
16 --
17 -- NOTE: momo.xml masks some phone numbers ("*****013") and shows
```

Text to DDL

Query SQL

```
1 -- Basic CRUD demonstration (Schema SQL panel re-seeds the DB before each
2 -- Run,
3 -- so CREATE/UPDATE/DELETE + verification READ must happen together in one
4 -- query batch)
5 -- CREATE
6 -- INSERT INTO users (full_name, phone_number, user_type) VALUES ('Grace
7 -- Uwimana', '250788123456', 'CUSTOMER');
8 -- UPDATE
9 -- UPDATE transactions SET amount = 1500.00 WHERE transaction_id = 2;
10 -- DELETE
11 -- DELETE FROM system_logs WHERE stage = 'PARSE' AND processing_status =
12 -- 'FAILED';
13 -- READ: verify all three writes landed
14 -- SELECT
15 -- (SELECT COUNT(*) FROM users WHERE full_name = 'Grace Uwimana')
```

Copy as Markdown

Results

There are no results to be displayed.

Query #2 Execution time: 0.62ms

There are no results to be displayed.

Query #3 Execution time: 0.56ms

There are no results to be displayed.

Query #4 Execution time: 0.38ms

grace_uwimana_created	txn2_amount_after_update	dead_letter_logs_remaining
1	1500.0	0

6. Unique Rules Enforced (security & accuracy)

Each rule below was actually **triggered against MySQL 8.0** (not just declared) — the screenshots below show the real rejection, confirming CHECK constraints are active on this engine version (they are silently ignored before MySQL 8.0.16).

#	Rule	Purpose	Observed error
1	UNIQUE (financial_transaction_id) on transactions	ETL idempotency — ingesting the same MoMo SMS backup must never duplicate a transaction	Duplicate entry '76662021700' for key 'transactions.financial_transaction_id'
2	CHECK (amount > 0)	An amount ≤ 0 signals a parsing bug upstream, not a real transaction	Check constraint 'chk_txn_amount_positive' is violated.
3	FOREIGN KEY (category_id) ... ON DELETE RESTRICT	Every transaction must resolve to a known category; blocks orphan inserts	Cannot add or update a child row: a foreign key constraint fails (...fk_txn_category...)
4	ON DELETE RESTRICT from transactions → transaction_categories	A category still referenced by transaction history cannot be deleted, protecting historical classification	Cannot delete or update a parent row: a foreign key constraint fails (...fk_txn_category...)
5	UNIQUE (transaction_id, role) on transaction_participants	A transaction cannot have two SENDERS, two AGENTS, etc.	Duplicate entry '1-SENDER' for key 'transaction_participants.uq_txn_id_role'

#	Rule	Purpose	Observed error
6	CHECK (fee >= 0)	Rejects an impossible negative fee from a bad parse	Check constraint 'chk_txn_fee_nonnegative' is violated.

CHECK (balance_after IS NULL OR balance_after >= 0) uses the identical chk_* mechanism verified in rules 2 and 6 above, so one working example of each mechanism (UNIQUE, CHECK, FK RESTRICT) is treated as sufficient proof it applies engine-wide; it was not screenshotted separately.

6.1 Screenshot evidence

Rule 1 — UNIQUE (financial_transaction_id)

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6 -- enforcement; on 5.7/8.0-8.0.15 CHECK clauses parse but are
7 -- ignored)
8 -- How to run : mysql -u <user> -p < database_setup.sql
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14 -- in a transaction. Populated from names + phone numbers found in
15 -- the SMS body text.
16 --
17 -- NOTE: momo.xml masks some phone numbers ("*****013") and shows
```

Text to DDL

Query SQL

```
1 -- UNIQUE constraint demo: financial_transaction_id is the MoMo idempotency
2 -- key
3 -- (the ETL upserts on this value so a re-ingested SMS never duplicates a
4 -- transaction).
5 -- Uses a fresh sms_id (12) so only the financial_transaction_id UNIQUE is
6 -- being tested.
7 -- INSERT INTO sms_messages (sms_id, address, body, sms_timestamp, raw_hash)
8 -- VALUES (12, 'M-Money', 'duplicate test SMS', '2024-06-01 00:00:00',
9 -- SHA2('dup-test', 256));
10
11 -- INSERT INTO transactions (sms_id, category_id, financial_transaction_id,
12 -- amount, direction, transaction_datetime)
13 -- VALUES (12, 1, '76662021700', 2000.00, 'CREDIT', '2024-06-01 00:00:00');
```

Results

Query Error: Duplicate entry '76662021700' for key 'transactions.financial_transaction_id'

Rule 2 — CHECK (amount > 0)

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Schema SQL

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4 --
5 -- Target engine : MySQL 8.0.16+ (required for CHECK constraint
  enforcement;
6 --           on 5.7/8.0.0-8.0.15 CHECK clauses parse but are
  ignored)
7 -- How to run   : mysql -u <user> -p < database_setup.sql
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14 --   in a transaction. Populated from names + phone numbers found in
15 --   the SMS body text.
16 --
17 -- NOTE: momo.xml masks some phone numbers ("*****013") and shows
```

Text to DDL

Query SQL

```
1 -- CHECK constraint demo: amount must be strictly positive.
2 -- A negative/zero amount here would signal a parsing bug upstream in
  etl/clean_normalize.py,
3 -- not a valid MoMo transaction, so the database rejects it outright.
4 INSERT INTO transactions (sms_id, category_id, amount, direction,
  transaction_datetime)
5 VALUES (12, 1, -500.00, 'DEBIT', '2024-06-01 00:00:00');
```

Results

Query Error: Check constraint 'chk_bxn_amount_positive' is violated.

Rule 3 — FOREIGN KEY (category_id)

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6 --           on 5.7/8.0.0-8.0.15 CHECK clauses parse but are
  ignored)
7 -- How to run   : mysql -u <user> -p < database_setup.sql
8 -- =====
9
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13 --   People or entities that appear as a party (sender/receiver/agent)
14 --   in a transaction. Populated from names + phone numbers found in
15 --   the SMS body text.
16 --
17 -- NOTE: momo.xml masks some phone numbers ("*****013") and shows
```

Text to DDL

Query SQL

```
1 -- FOREIGN KEY constraint demo: category_id must reference a real
  transaction_categories row.
2 -- Guards against a bug in etl/categorize.py emitting an unrecognized
  category code.
3 INSERT INTO sms_messages (sms_id, address, body, sms_timestamp, raw_hash)
4 VALUES (12, 'M-Money', 'fk test SMS', '2024-06-01 00:00:00', SHA2('fk-test',
  256));
5
6 INSERT INTO transactions (sms_id, category_id, amount, direction,
  transaction_datetime)
7 VALUES (12, 999, 500.00, 'DEBIT', '2024-06-01 00:00:00');
```

Results

Query Error: Cannot add or update a child row: a foreign key constraint fails ('test`.`transactions`, CONSTRAINT 'fk_bxn_category' FOREIGN KEY ('category_id') REFERENCES 'transaction_categories' ('category_id') ON DELETE RESTRICT ON UPDATE CASCADE)

Rule 4 — ON DELETE RESTRICT

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4 --
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7 -- How to run   : mysql -u <user> -p < database_setup.sql
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9
10
11 -- -----
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13 --   People or entities that appear as a party (sender/receiver/agent)
14 --   in a transaction. Populated from names + phone numbers found in
15 --   the SMS body text.
16 --
17 --   NOTE: momo.xml masks some phone numbers ("*****013") and shows
```

Text to DDL

Query SQL

```
1 |-- ON DELETE RESTRICT demo: a category still referenced by transactions
  cannot be deleted.
2 -- Protects transaction history from silently losing its category
  classification.
3 DELETE FROM transaction_categories WHERE category_id = 1;
```

Results

Query Error: Cannot delete or update a parent row: a foreign key constraint fails ('test','transactions', CONSTRAINT 'fk_txn_category' FOREIGN KEY ('category_id') REFERENCES 'transaction_categories' ('category_id') ON DELETE RESTRICT ON UPDATE CASCADE)

Fiddle Title

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Fiddle Description

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SQL Assessments
in minutes

Schema SQLQuery SC

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Text to DDL

Results

Query Error:

Duplicate entry '1-SENDER' for key 'transaction_particip

Rule 6 — CHECK (fee >= 0)

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Fiddle Description

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SQL Assessments
in minutes

Schema SQLQuery SC

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Text
to
DDL

Results

Query Error:
Check
constraint
'chk_txn_fee_nonne'
is violated.

7. Mapping Back to the Repository

Deliverable	File
ERD (editable)	docs/erd_diagram.drawio
ERD (image)	docs/erd_diagram.png
SQL DDL + DML	database/database_setup.sql
JSON schemas	examples/json_schemas.json
This document	docs/database_design.md
Verification screenshots	docs/screenshots/01-08_*.png